

Chemistry in Times of Artificial Intelligence

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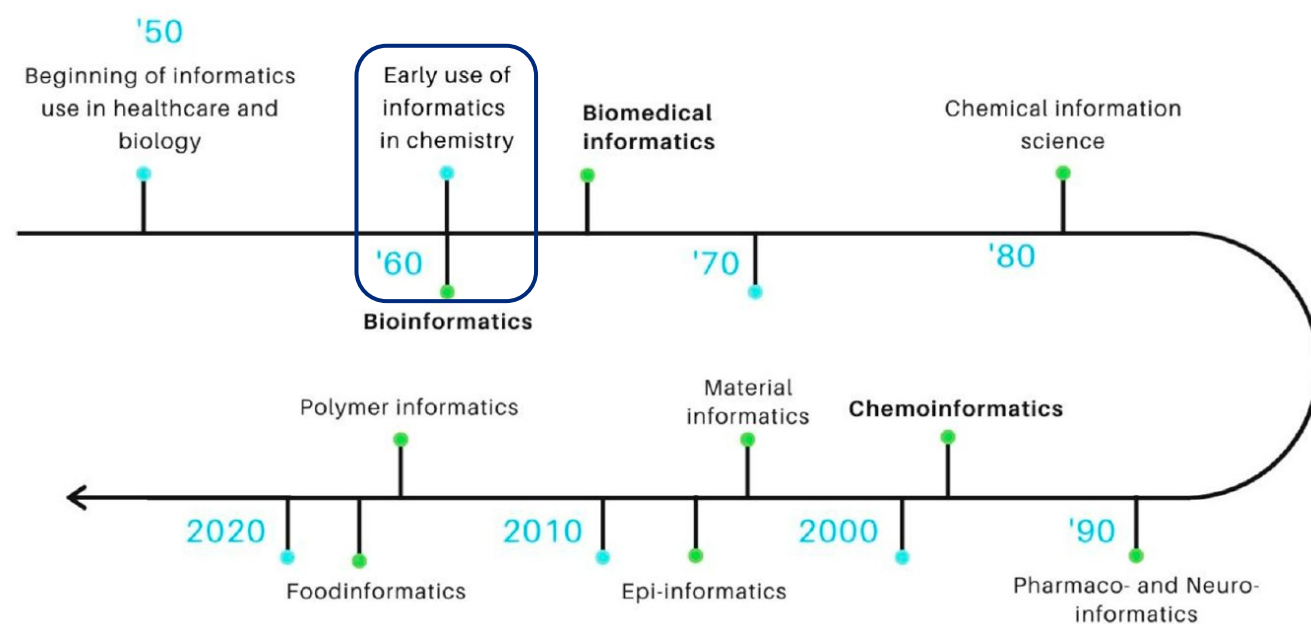
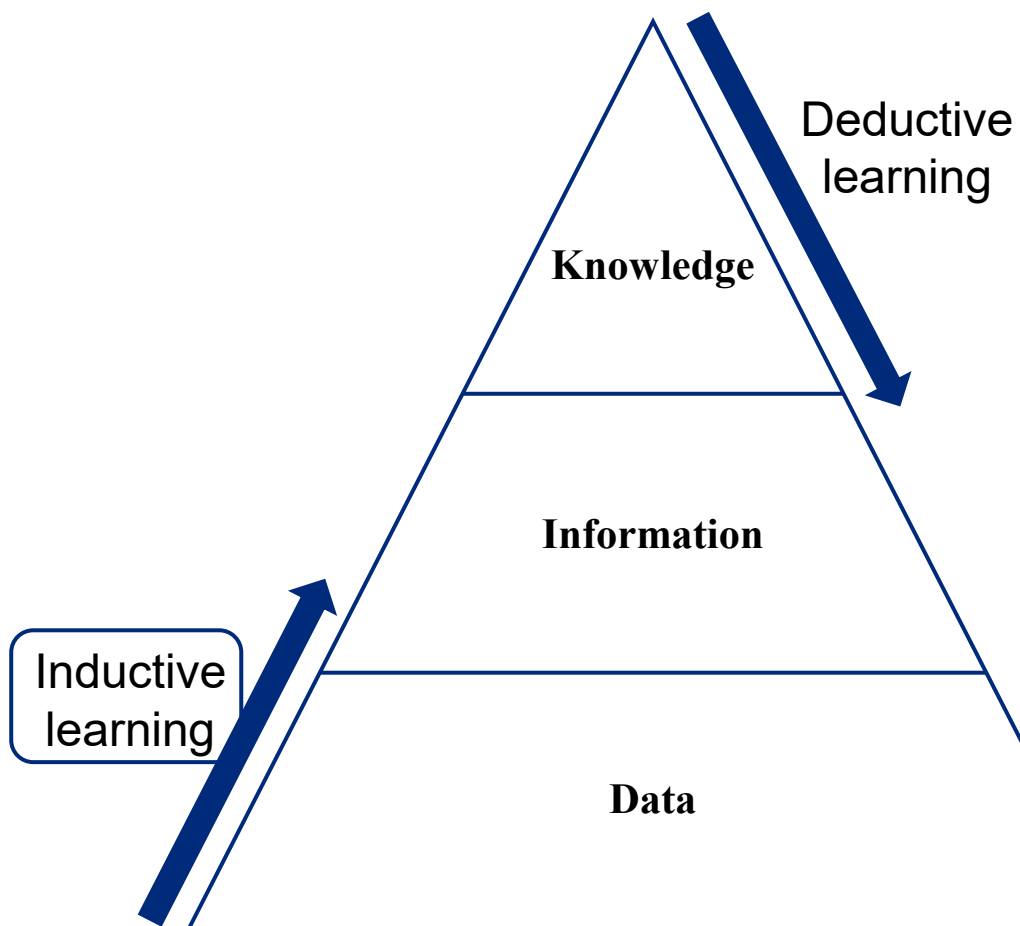
- J. Gasteiger, Chemistry in Times of Artificial Intelligence. ChemPhysChem 2020, 21, 2233.
- Chemoinformatics and artificial intelligence colloquium: progress and challenges in developing bioactive compounds.

Chemistry in Times of Artificial Intelligence

- Inductive and deductive learning in chemistry
- The beginning
- DENDRAL project
- Some achievements in cheminformatics
- The “black box” AI problem

Reflections

Inductive and deductive learning in chemistry



“The study of processes for storing and obtaining information”

The beginning

QSAR/QSPR

p - σ - π Analysis. A Method for the Correlation of Biological Activity and Chemical Structure

[Corwin. Hansch](#); [Toshio. Fujita](#)

+ [Author & Article Information](#)

Journal American Chemical Society (1964) 86 (8): 1616–1626.

Synthesis

Computer-Assisted Design of Complex Organic Syntheses:

Pathways for molecular synthesis can be devised with a computer and equipment for graphical communication.

[E. J. COREY AND W. TODD WIPKE](#) [Authors Info & Affiliations](#)

SCIENCE • 10 Oct 1969 • Vol 166, Issue 3902 • pp. 178-192 • DOI: [10.1126/science.166.3902.178](#)

Structure elucidation

Automated structure elucidation of several kinds of aliphatic and alicyclic compounds

[Shinichi. Sasaki](#); [Hidetsugu. Abe](#); [Tatsumi. Ouki](#); [Masayoshi. Sakamoto](#); [Shukichi. Ochiai](#)

+ [Author & Article Information](#)

Anal. Chem. (1968) 40 (14): 2220–2223.

- Corwin. Hansch, Toshio. Fujita; p - σ - π Analysis. A Method for the Correlation of Biological Activity and Chemical Structure. *Journal American Chemical Society* 1 April 1964; 86 (8): 1616–1626.
- E. J. Corey, W. Todd Wipke, Computer-Assisted Design of Complex Organic Syntheses. *Science* 166, 178-192 (1969).
- Shinichi. Sasaki, Hidetsugu. Abe, Tatsumi. Ouki, Masayoshi. Sakamoto, Shukichi. Ochiai; Automated structure elucidation of several kinds of aliphatic and alicyclic compounds. *Anal. Chem.* 1 December 1968; 40 (14): 2220–2223.

DENDRAL Project

“a computer program designed to infer molecular structure from mass spectrometry data”

-Edward Feigenbaum, Bruce Buchanan, and Joshua Lederberg at Stanford University (1965)-

Present



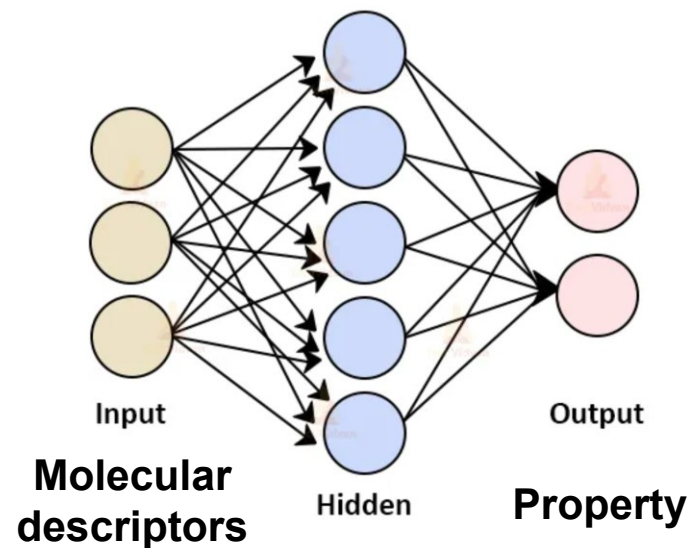
...to run the computationally intensive quantum chemistry simulations integrated into ChemGraph. (Image by Argonne National Laboratory)

Some achievements in cheminformatics

Databases



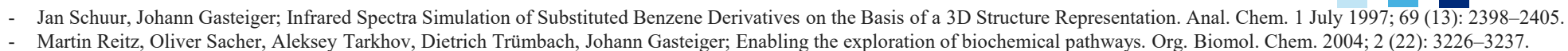
Prediction of properties



Analytical chemistry

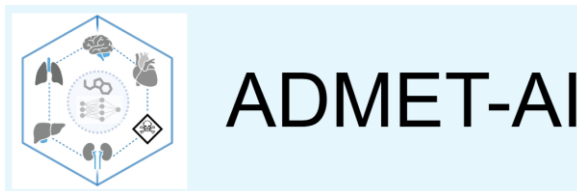
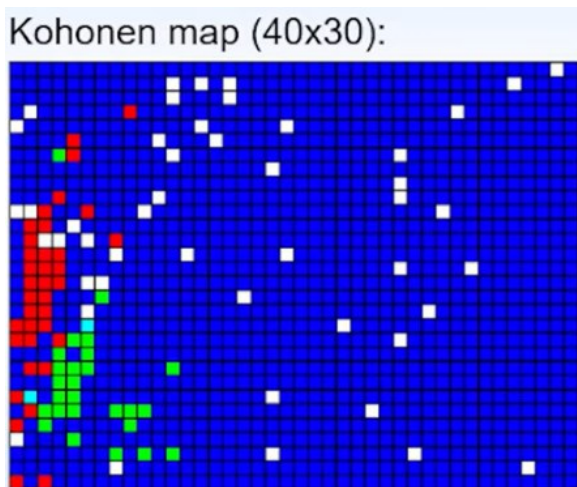


Biochemical pathways



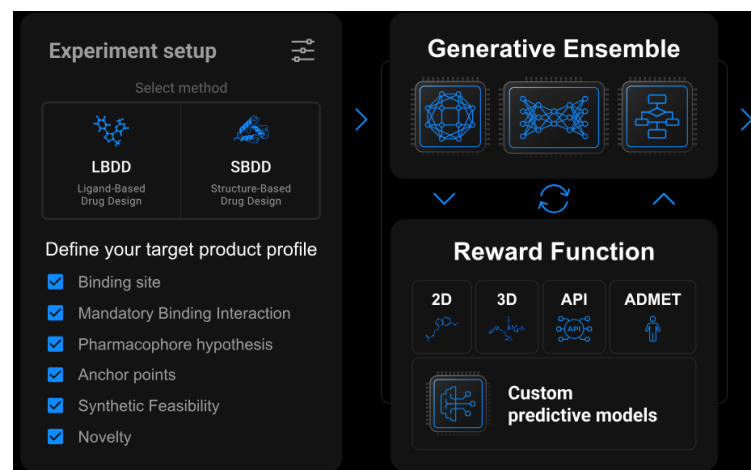
Some achievements in cheminformatics

Drug discovery



**Insilico
Medicine**

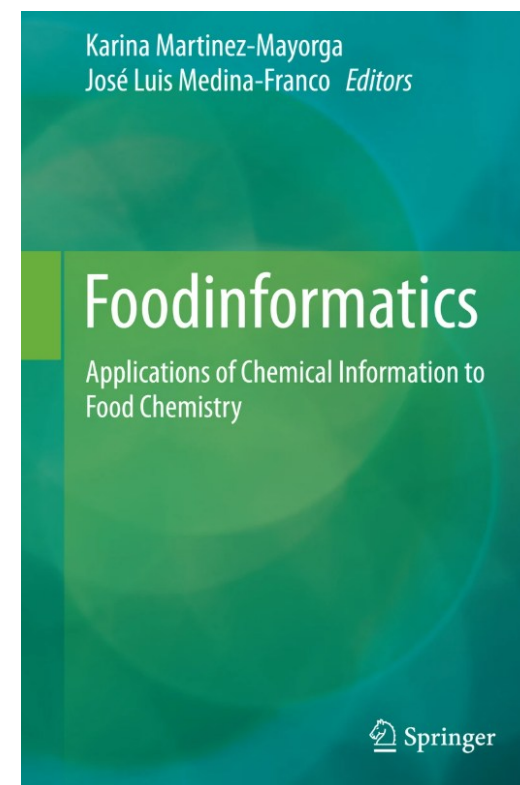
Agricultural



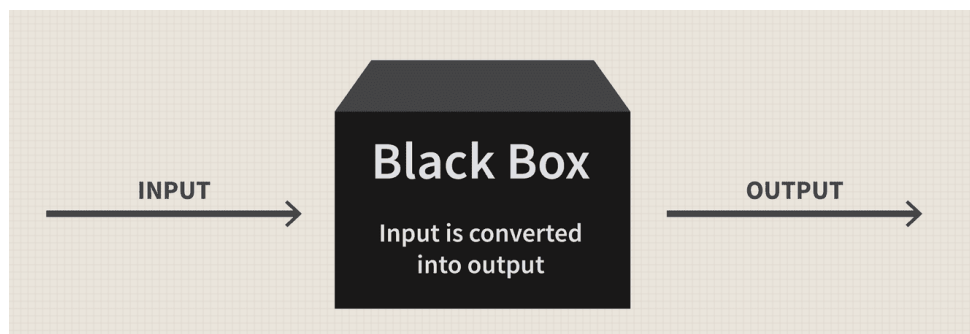
syngenta



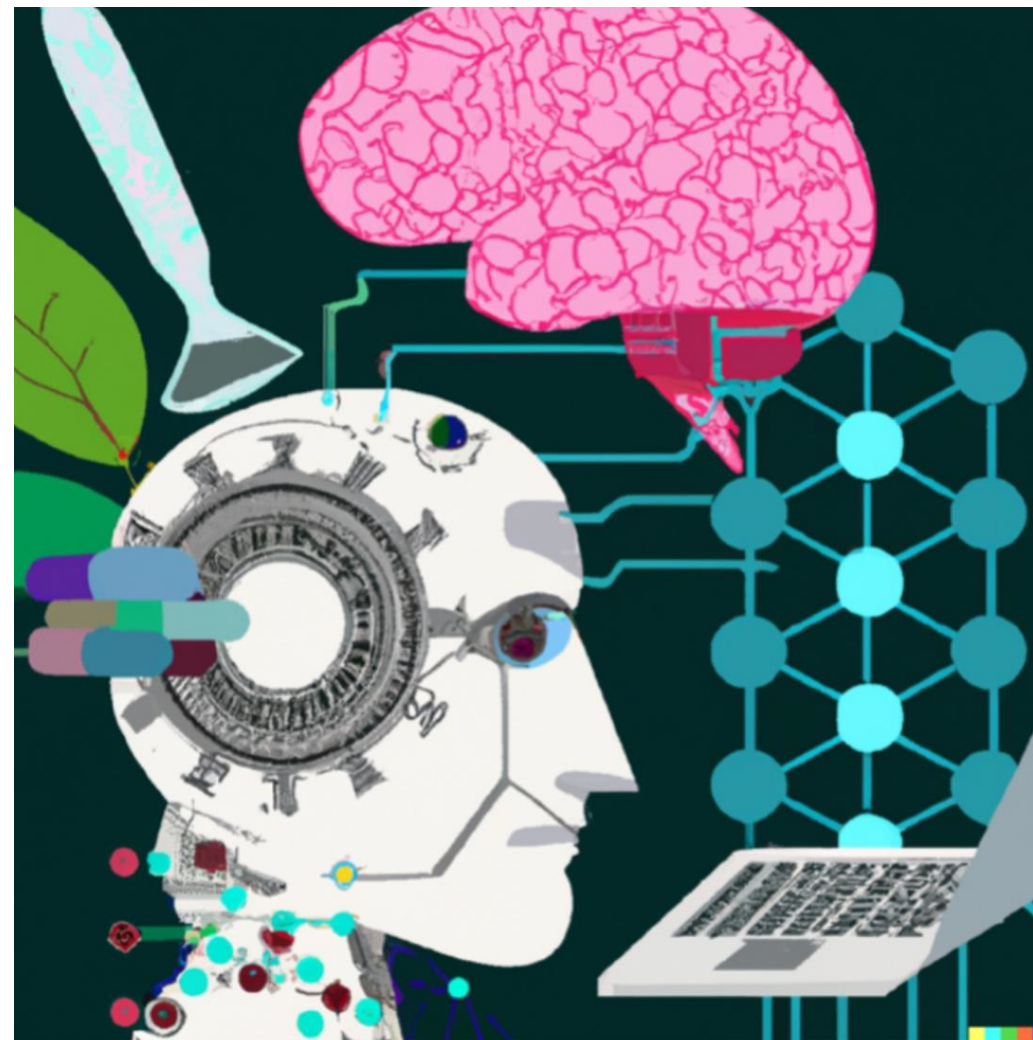
Food science



The “black box” AI problem



One of the critical limitations of current AI systems is the “black box” problem, where model predictions are difficult to interpret.



Reflections

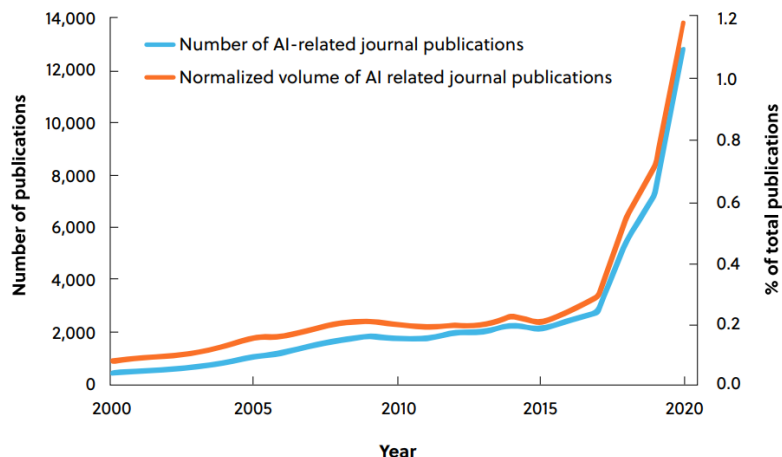
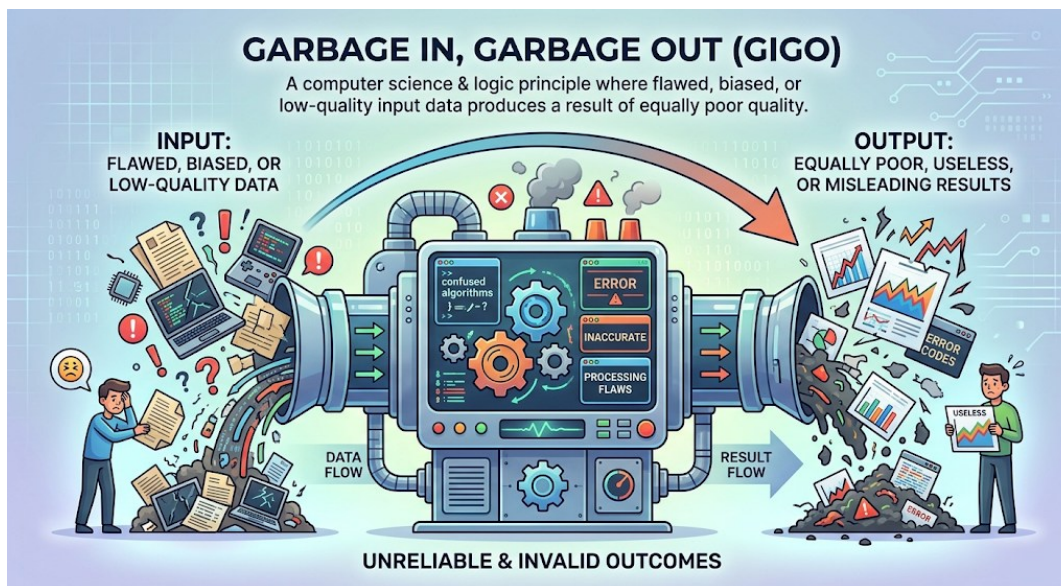
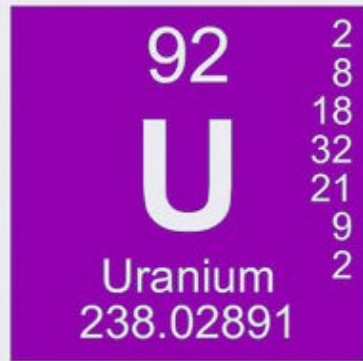
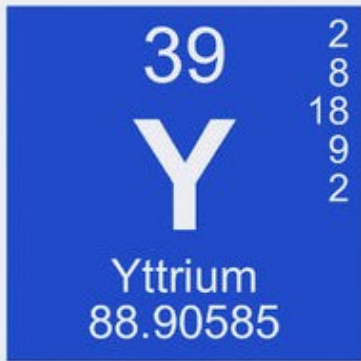
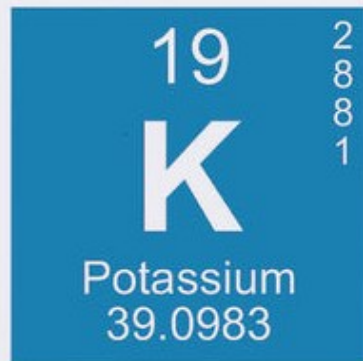
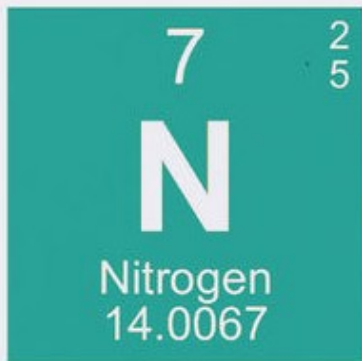
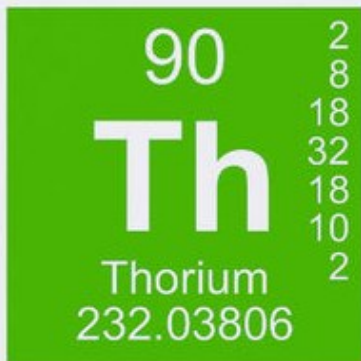


Figure 2. Annual publication volume of AI-related chemistry journal articles from 2000 to 2020



- Since the 1960s, we have been gaining knowledge from data.
- Computing power does not replace scientific judgment or human reasoning in the interpretation of results.
- Avoid using artificial intelligence or machine learning just to follow trends. Use them with a clear methodological purpose.
- Garbage in – Garbage out !!!



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ANY QUESTIONS?

